

REVUB: User Manual (version 1.2)

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1 The REVUB model (“Renewable Electricity Variability, Upscaling and Balancing”) was developed
2 specifically to address the challenge of optimising hydro-solar-wind complementarity through smart
3 hydropower operation and smart choice of solar-wind portfolios. We first explain the general principles
4 of the REVUB model (Note 1), based on which the different modules of the model are explained one
5 by one: hydropower plant classification (Note 2); simulation of hydropower generation and reservoir
6 dynamics (Note 3); and simulation and optimisation of joint hydro-solar-wind operation (Note 3 to
7 Note 5). An optional, additional assessment of pumped-storage potential with REVUB is described
8 in Note 6. The possibilities of modelling interactions between hydropower plants in cascades are
9 described in Note 7. Some notes on spinning reserve requirements are summarised in Note 8. An
10 overview table of terms and symbols used in the model description is given in Note 9.
11 The most recent version of the REVUB code was written for Python environments and can be accessed
12 via <https://github.com/VUB-HYDR/REVUB> and used under the MIT license. All equations given in
13 this document are referenced in the code wherever relevant/used. The ways to prepare input, run
14 the code, and interpret output are given in Note 10, Note 11 and Note 12. The manual ends with an
15 overview of additional modelling tips in Note 13.

16 **Note 1 Principles of REVUB**

17 The basic idea of REVUB is to assess the potential of reservoir hydropower to assist in the grid inte-
18 gration of variable power generation, e.g. from solar PV and wind power. Since reservoir hydropower
19 can be flexibly dispatched, the premise is that every hydropower plant can help compensate for part
20 of the variability of the pooled solar/wind resources on the same grid^{1,2}. This requires adaptations,
21 down to hourly resolution, of the conventional reservoir rule curves applied to keep reservoir water
22 levels within acceptable ranges on seasonal and multiannual time scales.
23 REVUB calculates the adapted rule curves required for balancing a given solar/wind mix, and op-
24 timises the amount of solar/wind power whose variability each hydropower plant can compensate,
25 based on three criteria: (i) reservoir lake levels and emptying-refilling must remain comparable under
26 the new rule curves to those under conventional reservoir operation, on seasonal and multiannual
27 time scales; (ii) the resulting hydro-solar-wind power mix must be reliable, i.e. capable of consistently
28 meeting a certain load from hour to hour, month to month and year to year without failure (un-
29 less, in rare cases, extreme droughts force hydropower curtailment); and (iii) downstream stable (e.g.
30 environmental and/or minimum loading) flow constraints are never to be violated. An overview of
31 the calculation steps performed by REVUB is given in Fig. S1. The steps outlined in this chart are
32 described in more detail in the next sections, as indicated.

33 **Note 2 Hydropower plant categorisation**

34 In REVUB, hydropower plants are categorised into two groups, based on whether the average natural
35 inflow would take more or less than one year to fill the reservoir; in other words, whether the reservoir
36 is “large” or “small” compared to the inflow. For plants in the “large” category, rule curves can be
37 designed to ensure outflow and power output are relatively stable from month to month^{3,4}, enabling
38 the plant to serve as baseload provider. Such rule curves typically boil down to parameterisations of
39 required outflow as a function of lake level⁵. For hydropower plants in the “small” category, rule curves
40 should additionally take into account that during the wet season(s), certain fractions of the received
41 water need to be directly released from the reservoir without being stored, as reservoir capacity would
42 be insufficient for this⁶. Small hydropower plants thus operate partly as run-of-river plants.
43 In mathematical terms, the classification of hydropower plants as “large” or “small” is done by calcu-
44 lating the dimensionless filling time τ_{fill} :

$$\tau_{fill} = \frac{V_{max}/T_{year}}{Q_{in,nat}(t)}, \quad (S1)$$

45 where V_{max} is the reservoir volume (in m³), T_{year} is the number of seconds in a year, and $Q_{in,nat}(t)$
46 denotes natural reservoir inflow (in m³/s). By default, hydropower plants are classified by REVUB

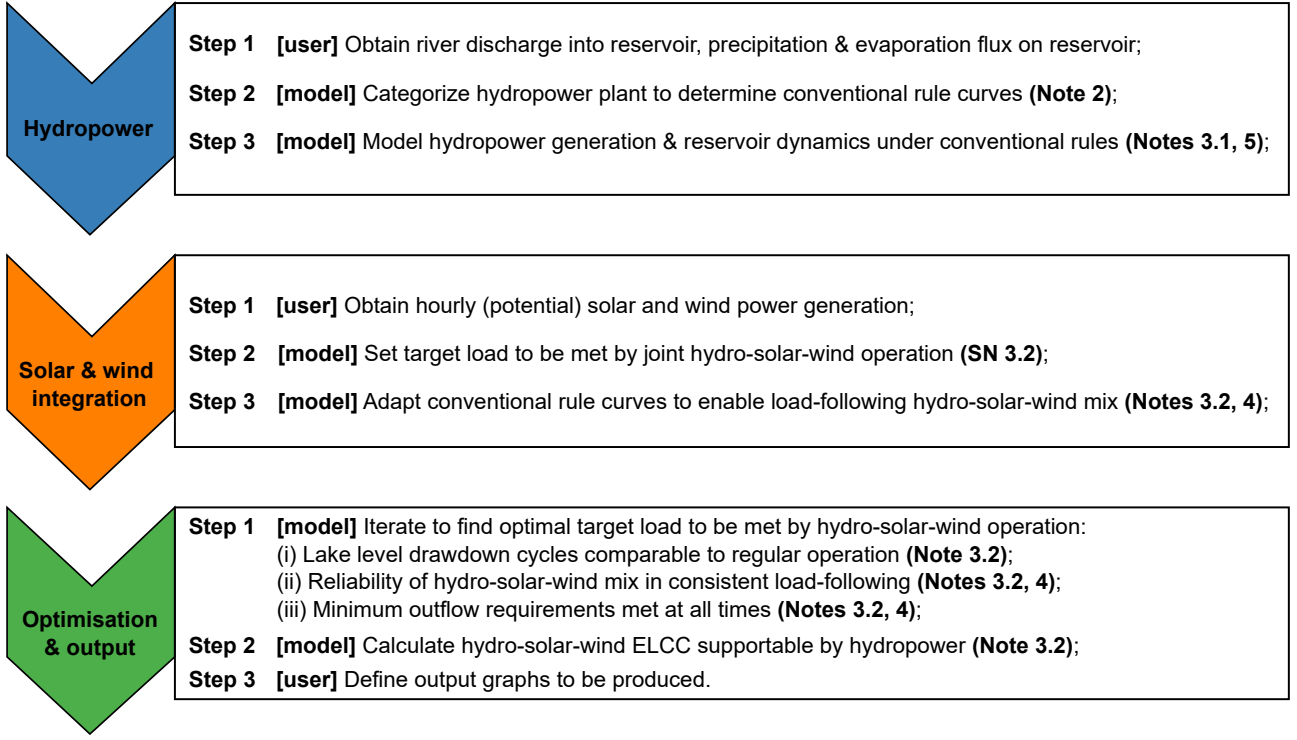


Figure S1: **Flowchart of the REVUB model approach**, explaining the steps taken in modelling hydro, solar and wind power and their integration. ELCC = Effective Load Carrying Capability. See also Note 10, Note 11 and Note 12 for the ways to prepare input, run the code, and interpret output.

as large or small based on whether their τ_{fill} is larger or smaller than unity, respectively; but this can be changed by the user.

Regardless of whether a plant is classified as large or small, it could be operated according to alternative rule curves with similar seasonal to multiannual outcomes as those resulting from conventional operation, but with additional patterns in water release from sub-daily to seasonal timescales, designed for balancing supply with demand in a power mix with a high share of variable renewable electricity (RE). This would mean allocating a portion of the inflowing water for flexible use and releasing it to generate extra power whenever needed to meet a certain load, such as during evening and night hours when solar PV output is low or zero.

Even for large hydropower plants, only part of the water intake can be allocated for flexible use, as a minimum stable outflow usually has to be guaranteed for at least four reasons (two technical reasons and two environmental reasons): (i) ensuring that turbines run in their high-efficiency range, which requires a certain minimum loading [*technical*]; (ii) safeguarding power system stability, since requirements for inertia impose having some synchronous capacity, such as from hydroturbine-driven generators, dispatched at all times⁷ [*technical*]; (iii) safeguarding downstream ecological integrity through a minimum environmental outflow⁸ [*environmental*]; and (iv) supplying irrigation water demand downstream [*environmental*].

Thus, regulated reservoir outflow must have a stable and a flexible component, which are denoted respectively $Q_{stable}(t)$ and $Q_{flexible}(t)$ (both in m^3/s). $Q_{stable}(t)$ can be further split up into a technical component $Q_{stable}^{tech}(t)$ and its environmental/irrigation component $Q_{stable}^{env/irr}(t)$. These components are not mutually exclusive, as one can contribute to the other; thus, in general, $Q_{stable}(t)$ is not the sum of both components, but simply the maximum value of both (the one that takes precedence over the other, thereby automatically fully covering the other).

Overall flexibility can be increased by reducing $Q_{stable}(t)$ in favour of $Q_{flexible}(t)$. Essentially, to allocate a certain amount of water for flexible use, the stable outflow will have to be reduced by a comparable amount such that the overall water budget remains unchanged. Since $Q_{stable}^{env/irr}(t)$ is usually

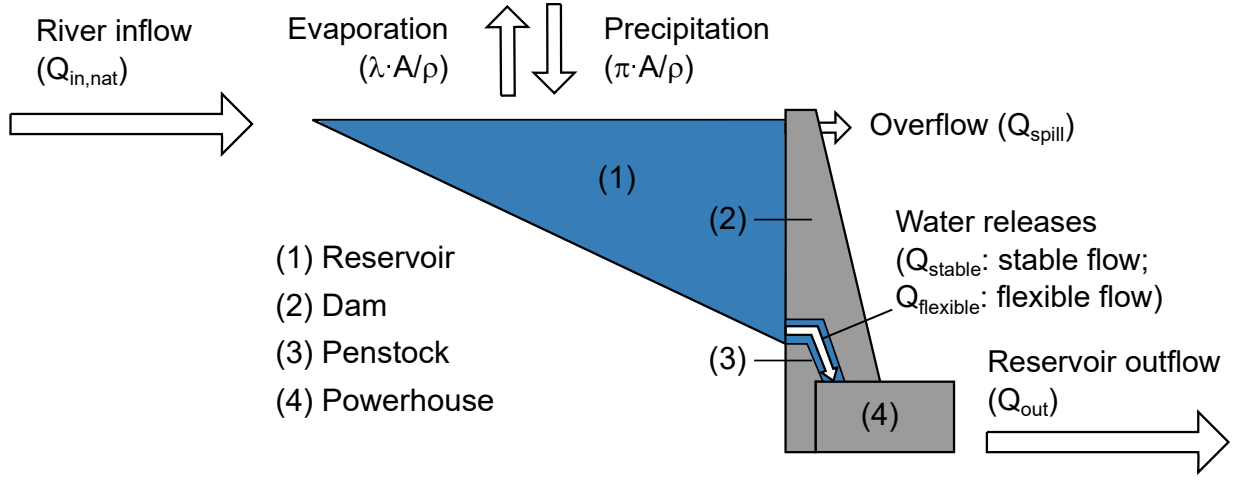


Figure S2: **Flowchart of the water gains and losses of a large hydropower plant as modelled in REVUB.** Gains consist of river discharge and lake surface precipitation; losses consist of reservoir outflows and lake surface evaporation.

prescribed externally, in practice, this means that allocating more water to flexibility implies reducing $Q_{stable}^{tech}(t)$. Total reservoir outflow may have two other components: a seasonal (“run-of-river”) component, denoted $Q_{RoR}(t)$, for small hydropower plants; and an overflow-prevention component⁵ $Q_{spill}(t)$, representing the release of water via spillways when filling levels approach critical values, e.g. during extremely wet periods. Thus, total reservoir outflow $Q_{out}(t)$ is, in general, given by

$$\begin{aligned} Q_{out}(t) &= Q_{stable}(t) + Q_{flexible}(t) + Q_{RoR}(t) + Q_{spill}(t) \\ &= \max \left[Q_{stable}^{tech}(t), Q_{stable}^{env/irr}(t) \right] + Q_{flexible}(t) + Q_{RoR}(t) + Q_{spill}(t). \end{aligned} \quad (S2)$$

for both large and small hydropower plants. Correspondingly, hydropower generation consists of components $P_{stable}^{hydro}(t)$, $P_{flexible}^{hydro}(t)$ and $P_{RoR}^{hydro}(t)$ (in MW)ⁱ. In the next sections, the simulations in REVUB of flexibility provision by large and small hydropower plants are discussed in detail. Note 3 and Note 4 deal with large plants (which are the exception rather than the rule; but starting with these is informative as the mathematics are somewhat simpler); Note 5 extends the principles to small plants.

Note 3 Reservoir simulation for large hydropower plants

The volume of water $V(t)$ contained in a large hydropower plant’s reservoir is given by the recursive relation

$$V(t + \Delta t) = V(t) + \left[Q_{in,nat}(t) - Q_{stable}(t) - Q_{flexible}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)] \frac{A(t)}{\rho} \right] \Delta t, \quad (S3)$$

where $\pi(t)$ and $\lambda(t)$ are respectively precipitation and evaporation flux ($\text{kg}/\text{m}^2/\text{s}$; see Methods); $A(t)$ is the surface area of the reservoir lake (m^2); ρ the density of water (taken to be $1000 \text{ kg}/\text{m}^3$); and Δt is the simulation time step (one hour is the default in REVUB). Seepage losses, which are usually very small in reservoirs^{6,9}, are neglected in the REVUB code (but could be included e.g. via a correction factor to total inflow). A schematic diagram of the involved water fluxes is given in Fig. S2.

Note 3.1 explains how $Q_{stable}(t)$ and the corresponding power generation $P_{stable}^{hydro}(t)$ are modelled for conventional, baseload-oriented operation, in which the flexible components are zero. Simulating conventional operation is necessary to have a reference time series of long-term lake volume fluctuations, based on which the balancing-oriented operation (dispatch) can be calibrated. Subsequently, Note 3.2

ⁱIt is assumed that overflow prevention releases are never used for power generation, to avoid damaging turbine equipment and because such high flows would normally exceed turbine capacity anyway.

explains how $Q_{stable}(t)$ and $Q_{flexible}(t)$, and the corresponding power generation components $P_{stable}^{hydro}(t)$ and $P_{flexible}^{hydro}(t)$, are modelled for balancing-oriented operation, i.e. dispatch runs.

Note 3.1 Conventional operation

We denote conventional, baseload-oriented operational rules with the abbreviation “CONV”. By definition, $Q_{flexible,CONV}(t)$ and $P_{flexible,CONV}^{hydro}(t)$ are zero. The default way of modelling $Q_{stable,CONV}(t)$ in REVUB is based on logarithmic-exponential release rules^{4,10}, which works well in case conventional operational rules are unknown, and can be determined on the basis of long-term average inflow $\overline{Q_{in,nat}(t)}$ ⁱⁱ. Under the assumption that environmental/irrigation outflow $Q_{stable}^{env/irr}(t)$ is prescribed at all times, approximate rules used in REVUB are:

$$Q_{stable,CONV}^{tech}(t) = \begin{cases} \overline{Q_{in,nat}(t)} \left[d_{min} + \ln \left(\kappa \left[\frac{V_{CONV}(t)}{V_{max}} \right]^\phi + 1 \right) \right], & \text{for } V_{CONV}(t)/V_{max} < f_{opt} \\ \overline{Q_{in,nat}(t)} \left[\exp \left(\gamma \left[\frac{V_{CONV}(t)}{V_{max}} - f_{opt} \right]^2 \right) \right], & \text{for } V_{CONV}(t)/V_{max} \geq f_{opt} \end{cases}, \quad (S4)$$

where d_{min} is the fraction of yearly average inflow required as minimum stable outflow; V_{max} is the maximum reservoir storage capacity; f_{opt} is the optimal filling fraction; and κ , ϕ and γ are constants. κ and ϕ are given⁴ by

$$\kappa = f_{opt}^{-\phi} [\exp(1 - d_{min}) - 1], \quad (S5)$$

and

$$\phi = \alpha \tau_{fill}^{1/2}. \quad (S6)$$

Recommended default values are $f_{opt} = 80\%$, $\alpha = 2/3$, and $\gamma = 10$ based on generalised reservoir operation rules^{4,10}; for d_{min} , values may be chosen based on requirements for minimum turbinised flow to prevent turbines running at low efficiency, and/or generalised environmental flow rules⁸ (the former will usually take precedence over the latter). It is possible (and arguably desirable when running scenarios for the first time) for the user to not specify a value for d_{min} , and instead provide a value of minimum required load f_{min} on one single turbine (expressed as fraction of maximum turbine throughflow), reflecting the lower bound of the turbine’s operating range, based on which REVUB will calculate and work with a default value of d_{min} ⁱⁱⁱ. All the preceding parameters can be changed by the user in the process of calibration to historical data.

The overflow prevention component $Q_{spill,CONV}$ is modelled as

$$Q_{spill,CONV}(t) = \begin{cases} 0, & \text{for } V_{CONV}(t)/V_{max} < f_{spill} \\ \max \left[0, \left(Q_{in,nat}(t) + [\pi(t) - \lambda(t)] \frac{A_{CONV}(t)}{\rho} \right) (1 + \mu) - Q_{stable,CONV}(t) \right], & \text{for } V_{CONV}(t)/V_{max} \geq f_{spill} \end{cases}, \quad (S7)$$

where f_{spill} is the filling fraction at which the overflow prevention starts^{iv}, and μ represents a small fraction such that lake levels are brought below critical levels $f_{spill}V_{max}$ as quickly as possible without releasing unnaturally high flows downstream; recommended values are $f_{spill} = 95\%$ and $\mu = 0.1$, but these can be changed by the user.

To simulate minimum drawdown levels when facing drought-like situations, one further rule is added to equation (S4): outflow, and with it hydropower production, is automatically curtailed^{5,11} if the volume levels $V_{CONV}(t)$ dip below critical levels $f_{stop}V_{max}$, and only restarted once volumes have recovered to $f_{restart}V_{max}$. The values f_{stop} and $f_{restart}$ must be specified by the user, ideally based on reservoir bathymetry (see Note 3.3).

ⁱⁱBy default, REVUB considers this average to be across the entire modelled period, but the user can alternatively select a specific part of the input period on which to base the average.

ⁱⁱⁱThe calculation is as follows: $d_{min}^{default} = (f_{min} Q_{turb}^{max} / N_{turbines}) / \overline{Q_{in,nat}(t)}$, where $N_{turbines}$ is the number of turbines (units) of the plant. The term $f_{min} Q_{turb}^{max} / N_{turbines}$ represents the minimum flow in m³/s required to keep one turbine running within its operating range.

^{iv}Note that every hydropower plant has a certain safety level, i.e. the dam is always somewhat higher than the maximum water level⁵, to mitigate flood risk. Because of this safety level, should the overflow procedure fail for any reason, the dam would not immediately overflow. However, flooding would occur along the shores of the lake, which is why it is always prudent to prevent straining the safety level.

129 The corresponding stable power output $P_{stable,CONV}^{hydro}(t)$ can then be calculated as

$$P_{stable,CONV}^{hydro}(t) = \min \left[Q_{stable,CONV}(t); Q_{turb}^{max} \right] \eta_{turb} \rho g h_{CONV}(t), \quad (S8)$$

130 where η_{turb} is the turbine efficiency (%)^v, g the gravitational acceleration (9.81 m/s²), and $h(t)$ the
 131 hydraulic head (m), i.e. the difference in water level between the headwater behind the dam and the
 132 tailwater at the turbines. The value of η_{turb} should be user-defined, ideally based on knowledge of the
 133 plant's turbines^{vi}. The calculation of $h(t)$ is explained in Note 3.3. Q_{turb}^{max} is the maximum turbine
 134 throughput (at which the power generating capacity of the turbine is fully used). If this value cannot
 135 be found, it can be approximated with $Q_{turb}^{max} \approx P_{turb}^r / (\eta_{turb} \rho g h_{max})$, where P_{turb}^r is the rated power
 136 capacity of the hydropower plant and h_{max} is the maximum hydraulic head.

137 With these rules, results are mostly independent of arbitrary initial conditions (convergence to the
 138 same time series happens typically within 2 simulation months). REVUB uses the initial condition
 139 $V_{CONV}(0) = f_{init} V_{max}$ (and the corresponding lake area and water level; see Note 3.3) where the user
 140 can specify f_{init} .

141 We note that, in case the user wishes to include their own conventional reservoir rules, this is possible
 142 by correspondingly adapting the default REVUB code by adapting the lines referring to the above
 143 equations.

144 **Note 3.2 Balancing-oriented operation (dispatch)**

145 Reservoir dispatch oriented towards balancing solar and wind power variability is denoted with the
 146 abbreviation “BAL”. For reservoirs of large hydropower plants, BAL operation is modelled as follows.
 147 First, the needs for sub-daily to seasonal dispatching patterns must be established. The load difference
 148 $P_d(t)$ between total inflexible power generation (stable hydropower, solar power, and wind power) and
 149 power demand is calculated as:

$$\begin{aligned} P_d(t) &= P_{stable,BAL}^{hydro}(t) + c_{solar} \cdot CF_{solar}(t) + c_{wind} \cdot CF_{wind}(t) - L(t) \\ &= P_{inflexible}(t) - L(t). \end{aligned} \quad (S9)$$

150 Here, c_{solar} and c_{wind} represent a certain amount of solar PV and wind power capacity, respectively
 151 (in MW), whose value is to be optimised by REVUB; $CF_{solar}(t)$ and $CF_{wind}(t)$ represent the solar
 152 and wind capacity factor for each time step, respectively; and $L(t)$ represents a certain target load
 153 profile that is to be followed^{vii}.

154 The time series $L(t)$, by default, is set as follows:

$$L(t) = P_{load} L_{norm}(t), \quad (S10)$$

155 with P_{load} a constant determining the average load to be followed and reflecting the constraints
 156 on overproduction, and $L_{norm}(t)$ the normalised ($\overline{L_{norm}(t)} = 1$) hour-to-hour load profile. P_{load} is
 157 calculated as

$$P_{load} = p_i(P_{inflexible}(t)), \quad (S11)$$

158 where $p_i(x)$ denotes the i^{th} percentile of a variable x . Here, i thus represents the percentile of $P_{inflexible}$
 159 not exceeding P_{load} ; in other words, the percentage of time during which RE generation should not

^vHydropower plants usually produce reactive power next to active power (i.e. the power factor is less than unity). For purposes of REVUB, representing the power factor can be done through this same efficiency factor. For instance, to represent a conversion efficiency of 95% (a turbine quantity) and a power factor of 80% (an electric circuit quantity), one could set $\eta = 95\% \times 80\% = 76\%$.

^{vi}The efficiency of hydroturbines depends on the effective turbined flow at each moment, but is typically above 80% in realistic ranges of the latter^{12,13}. As explained above, the value d_{min} should at least reflect the fact that at any given moment, at least one of the $N_{turbines}$ turbines of a given plant must be active and running at a certain minimum partial load to allow consistent operation at high efficiency.

^{vii}Note that for the terms in equation (S9), MW and MWh are interchangeable units, due to the hourly time step employed in REVUB. Similarly, CF could be interpreted both as actual power generation divided by maximum potential generation in each hour (dimensionless fraction), or as power generation per unit of capacity (in hours).

160 exceed the average load to be followed. Generally, the lower i , the higher the amount of allowed excess
 161 production (overproduction) as compared to $L(t)$.
 162 In REVUB, the time series $CF_{solar}(t)$ and $CF_{wind}(t)$ must be provided as model input by the user.
 163 These could represent power generation from single locations, or weighted averages across the locations
 164 for on-grid solar or wind power selected for the region under scrutiny, to simulate the feed-in of solar
 165 and wind power from various locations into the same power grid. In case of the latter, we recommend
 166 assuming that the total capacity is distributed across locations within the region according to site-
 167 specific weight factors w_{solar}^n and w_{wind}^n . Then, $CF_{solar}(t)$ and $CF_{wind}(t)$ would be given by

$$CF_{solar}(t) = \frac{\sum_{n=1}^{N_{solar}} CF_{solar}^n(t) w_{solar}^n}{\sum_{n=1}^{N_{solar}} w_{solar}^n}; \quad CF_{wind}(t) = \frac{\sum_{n=1}^{N_{wind}} CF_{wind}^n(t) w_{wind}^n}{\sum_{n=1}^{N_{wind}} w_{wind}^n}, \quad (S12)$$

168 with $CF_{solar}^n(t)$ and $CF_{wind}^n(t)$ the capacity factors of solar PV or wind power for each time step,
 169 respectively, at each individual site with index n ; and N_{solar} and N_{wind} the number of sites for solar PV
 170 or wind power generation, respectively. The weight factors w_{solar}^n and w_{wind}^n represent preferences for
 171 certain sites over others for the development of solar PV and wind parks. To determine the site-specific
 172 weight factors, one could take e.g. the available area for deployment in each different site and assume
 173 it scales linearly with potential capacity deployment, as per ref.¹⁴. Another approach is that of ref.¹⁵,
 174 which took a site's weight factor to be equal to the multi-year average capacity factor for solar PV or
 175 wind power at that site, reflecting a preference for sites with higher yield:

$$w_{solar}^n = \overline{CF_{solar}^n(t)}; \quad w_{wind}^n = \overline{CF_{wind}^n(t)}. \quad (S13)$$

176 The capacity ratio $c_{solar} : c_{wind}$ should also be user-defined; it represents the relative share of solar
 177 and wind capacity to be deployed by the model. Sensitivity tests to find an optimal ratio can then be
 178 performed by running the model for several values of the capacity ratio.

179 Second, in BAL operation, $Q_{stable}(t)$ is reduced in favour of $Q_{flexible}(t)$ such that $L(t)$ can be met
 180 by the combination of stable hydropower, flexible hydropower, solar power and wind power. In the
 181 REVUB code, the default approach (which could be adapted by the user in the code) is to take
 182 $Q_{stable,BAL}^{tech}(t)$ as a fraction of the average inflow^{viii}:

$$Q_{stable,BAL}^{tech}(t) = (1 - C_{OR}) \overline{Q_{in,nat}(t)}. \quad (S14)$$

183 Here, the fraction C_{OR} denotes the “operating reserve coefficient”¹⁶, and determines the amount of
 184 water available for flexible use. Theoretically, as $C_{OR} \rightarrow 1$, the hydropower plant in question would
 185 operate near-completely flexibly and provide almost no minimum stable load, beyond the “coinci-
 186 dental” one from the environmental and irrigation outflow. This would also imply that the outflow
 187 of the dam could become extremely low for prolonged periods; for instance, when solar/wind power
 188 generation is high and/or demand is low. Such operation would likely violate inertia requirements
 189 and minimum load constraints of the plant. As default, REVUB therefore uses an upper bound of
 190 $C_{OR}^{max} = 1 - d_{min}$, such that the minimum outflow under BAL is always consistent with that under
 191 CONV. The selection of the optimal C_{OR} for each hydropower plant is explained in Note 4.

192 $P_{stable,BAL}^{hydro}(t)$ is calculated analogously to $P_{stable,CONV}^{hydro}(t)$ in equation (S8):

$$P_{stable,BAL}^{hydro}(t) = \min \left[Q_{stable,BAL}(t); Q_{turb}^{max} \right] \eta_{turb} \rho g h_{BAL}(t). \quad (S15)$$

193 Third, the necessary amount of flexible outflow $Q_{flexible,BAL}(t)$ and corresponding flexibly produced
 194 power $P_{flexible,BAL}^{hydro}(t)$ are determined for the instances when $P_{inflexible}(t)$ cannot meet $L(t)$, i.e. when
 195 $P_d(t) < 0$. The following rule then applies for $P_{flexible,BAL}^{hydro}(t)$ to maximise the followed load under
 196 peak capacity constraints and ramping constraints¹⁷:

$$P_{flexible,BAL}^{hydro}(t) = \begin{cases} 0, & \text{for } P_d(t) \geq 0 \\ \min \left[Q_{turb,flexible}^{pot}(t) \eta_{turb} \rho g h_{BAL}(t); \min \left[|P_d(t)|; P_{flexible,BAL}^{hydro}(t-1) + \Delta P_{turb}^{ramp} \right] \right], & \text{for } P_d(t) < 0 \text{ \& } \Delta P_d(t) < 0 \\ \min \left[Q_{turb,flexible}^{pot}(t) \eta_{turb} \rho g h_{BAL}(t); \max \left[|P_d(t)|; P_{flexible,BAL}^{hydro}(t-1) - \Delta P_{turb}^{ramp} \right] \right], & \text{for } P_d(t) < 0 \text{ \& } \Delta P_d(t) \geq 0 \end{cases}, \quad (S16)$$

^{viii}As for the CONV rules, by default this average is taken across the entire modelling period, but the user can alternatively select a part of that period as basis for the average.

where $\Delta P_d(t) = P_d(t) - P_d(t-1)$ determines whether the hydropower plant should ramp up ($\Delta P_d(t) < 0$) or down ($\Delta P_d(t) \geq 0$); ΔP_{turb}^{ramp} is the maximum ramp rate of the plant (in MW per time step); and

$$Q_{turb,flexible}^{pot}(t) = \max \left[0; Q_{turb}^{max} - Q_{stable,BAL}(t); \frac{P_{turb}^r - P_{stable,BAL}^{hydro}}{\eta_{turb} \rho g h_{BAL}(t)} \right]. \quad (S17)$$

This gives a flexible outflow

$$Q_{flexible,BAL}(t) = \frac{P_{flexible,BAL}^{hydro}(t)}{\eta_{turb} \rho g h_{BAL}(t)}. \quad (S18)$$

Fourth, the overflow prevention component $Q_{spill,BAL}(t)$ is calculated analogously to equation (S7):

$$Q_{spill,BAL}(t) = \begin{cases} 0, & \text{for } V_{BAL}(t)/V_{max} < f_{spill} \\ \max \left[0, \left(Q_{in,nat}(t) + [\pi(t) - \lambda(t)] \frac{A_{BAL}(t)}{\rho} \right) (1 + \mu) - Q_{stable,BAL}(t) - Q_{flexible,BAL}(t) \right], & \text{for } V_{BAL}(t)/V_{max} \geq f_{spill} \end{cases} \quad (S19)$$

As in CONV, to ensure that lake levels do not dip precariously low, $Q_{stable,BAL}$ and $Q_{flexible,BAL}$ are reduced to zero when $V_{BAL}(t) < f_{stop} V_{max}$ and restarted once $V_{BAL}(t) \geq f_{restart} V_{max}$ ^{ix}.

Since one needs the hydraulic head $h_{BAL}(t)$ to calculate $P_{stable,BAL}^{hydro}(t)$ from equation (S15), but $h_{BAL}(t)$ follows only from the overall operation, which requires knowledge of $P_{stable,BAL}^{hydro}(t)$ to calculate $P_d(t)$ according to equation (S9), this problem is solved iteratively by REVUB with an initial guess of $P_{stable,BAL}^{hydro}(t) = (1 - C_{OR}) P_{stable,CONV}^{hydro}(t)$, repeated until convergence (typically, 2-3 iterations suffice). The calculation of $h(t)$ is the same as for CONV and is explained in Note 3.3.

At given C_{OR} and given ratio $c_{solar} : c_{wind}$, REVUB optimises the above operation by identifying the time series $L(t) = L_{opt}(t)$, and the corresponding optimal $c_{solar} = c_{solar}^{opt}$ and $c_{wind} = c_{wind}^{opt}$, for which the resulting lake level time series is most comparable to what it would be under CONV operation. This optimisation is performed by finding the minimum of the relative deviation Ψ between the CONV and BAL lake levels across all time steps of an entire simulation period^x, as a function of c_{solar} and c_{wind} :

$$\Psi(c_{solar}, c_{wind}) = \frac{|(V_{BAL}(t) - V_{CONV}(t))|}{V_{CONV}(t)}, \quad (S20)$$

where

$$\min(\Psi) = \Psi(c_{solar}^{opt}, c_{wind}^{opt}). \quad (S21)$$

Note that the code could also be adapted to optimise lake levels compared to a certain mathematical rule curve, instead of to the outcome of “conventional” operational rules (Note 3.1). This would simply require exchanging $V_{CONV}(t)$ in equation (S20) by the corresponding rule curve of lake volume. Note furthermore that even when doing this, it is still useful to calculate the outcomes (hydropower generation and lake levels) resulting from conventional reservoir management, to verify (i) how well the rule curve can be followed *in general* under the given hydroclimate and occurrence of wet/dry years, and (ii) the extent to which the outcomes of flexible operation would differ from baseload-oriented operation.

REVUB allows the user the option to ensure that power droughts resulting from $V(t)$ dipping lower than $f_{stop} V_{max}$ can never be more severe under BAL operation than under CONV, by automatically discarding any solutions to equation (S21) where this would be the case.

Once the optimal solution is found, the Effective Load Carrying Capability (ELCC) of the hydro-solar-wind mix is calculated as follows. The maximum followable load $L_{followed}(t) = P_{followed} L_{norm}(t)$, which the hydro-solar-wind mix can meet without any load loss, is identified: this is the load at which the residual load ($L_{res}(t)$; the load minus the generation) has a maximum of zero. Thus, $P_{followed}$ is the value for which $\max[L_{res}(t)] = 0$, with $L_{res}(t)$ defined as:

$$L_{res}(t) = P_{followed} L_{norm}(t) - \left[P_{stable,BAL}^{hydro}(t) + P_{flexible,BAL}^{hydro}(t) + c_{solar}^{opt} \cdot CF_{solar}(t) + c_{wind}^{opt} \cdot CF_{wind}(t) \right]. \quad (S22)$$

^{ix}The length of periods when hydropower production is forced to stop, is important for due diligence planning and reflects in parameters such as the $P_{guaranteed}^{90}$ power (MW), i.e. the power that is guaranteed 90% of the time. Integration of VRE will generally improve such parameters, since VRE can continue producing even when hydropower is temporarily stopped and despite the fact that the flexibility from hydropower is lost in those periods. REVUB calculates $P_{guaranteed}^{xx}$ for a user-defined percentile xx for both CONV and BAL scenarios to showcase this.

^xOr a user-selected part of that simulation period, in line with the CONV and BAL rules.

232 The ELCC is then defined as the integral of $L_{followed}(t)$ over all time steps in a simulation year:

$$ELCC = \int_{year} L_{followed}(t) dt. \quad (S23)$$

233 The ELCC thus represents the total yearly load followed by each hydropower plant in combination with
 234 the solar and wind power whose variability it can compensate, under the optimal solution. The aggregate
 235 of ELCC across all hydropower plants represents the total followable load when all hydropower
 236 plants optimally contribute to compensating solar-wind variability; it is abbreviated $ELCC_{tot}$ here-
 237 after.

238 Ideally, $L_{followed}(t)$ should be equal to $L_{opt}(t)$ (and it usually is), but due to peaking constraints of
 239 hydropower plants (eq. (S17)), this is not guaranteed in every case. When $L_{followed}(t) < L_{opt}(t)$, the
 240 power plant's peaking capabilities are insufficient to meet all required peaks in $L_{opt}(t)$ together with
 241 solar and wind power. In such cases, the hydropower plant should run at somewhat lower flexibility,
 242 i.e. lower C_{OR} , at which a lower $L_{opt}(t)$ will apply that would put less strain on the plant's peaking
 243 capabilities, ensuring that $L_{followed}(t) \approx L_{opt}(t)$. This resimulation is done automatically by the
 244 REVUB code (Note 4).

245 The total contributions of hydro, solar and wind power to yearly electricity generation, denoted
 246 respectively $E_{reservoir}^{hydro}$, E_{solar} and E_{wind} , are obtained by integrating their respective power output
 247 across all time steps in a simulation year. First, $E_{reservoir}^{hydro}$ is equal to

$$E_{reservoir}^{hydro} = E_{stable}^{hydro} + E_{flexible}^{hydro} = \int_{year} (P_{stable}^{hydro}(t) + P_{flexible}^{hydro}(t)) dt. \quad (S24)$$

248 Second, E_{solar} and E_{wind} are obtained by multiplying c_{solar}^{opt} and c_{wind}^{opt} by $CF_{solar}(t)$ and $CF_{wind}(t)$,
 249 respectively, and integrating:

$$\begin{aligned} E_{solar} + E_{wind} &= \int_{year} (P_{solar}(t) + P_{wind}(t)) dt \\ &= \int_{year} (c_{solar}^{opt} \cdot CF_{solar}(t) + c_{wind}^{opt} \cdot CF_{wind}(t)) dt. \end{aligned} \quad (S25)$$

250 **Note 3.3 Head-volume-area relationships**

251 Calculating the hydraulic head $h(t)$ for each time step requires knowing the bathymetry and volume-
 252 area-depth relationship of each reservoir. Typical curves approximating such relationships need to be
 253 calibrated on a reservoir-to-reservoir basis. The user should provide bathymetric curves as input to
 254 each simulation. In case these are not available, we recommend using an archetypal reservoir shape
 255 function^{18,19}, modelling reservoirs' area-volume relationships with the equation

$$A(t) = A_{max} \left[\frac{V(t)}{V_{max}} \right]^{(1 - C_{reservoir})}, \quad (S26)$$

256 where $C_{reservoir} = V_{max}/(A_{max}z_{max})$ is the "reservoir coefficient", and A_{max} and z_{max} represent the
 257 maximum reservoir lake area and depth, respectively.

258 For any $V(t + \Delta t)$ calculated using equation (S3), the corresponding $A(t + \Delta t)$ can then be calculated
 259 using equation (S26). To obtain the hydraulic head $h(t + \Delta t)$ at each time step, the incremental
 260 change $\Delta h(t)$ is added to the head $h(t)$ of the previous time step, using the first-order approximation

$$\Delta h(t) \approx \frac{V(t+1) - V(t)}{A(t)}. \quad (S27)$$

261 When using equation (S26), before any simulation according to CONV or BAL in REVUB, a dummy
 262 simulation spanning all possible values of V , and thus of A and h , should be run to obtain a calibration
 263 (h, V) curve. In CONV and BAL simulations, the head $h(t + \Delta t)$ can then always be obtained from
 264 this calibration curve after $V(t + \Delta t)$ has been determined from equation (S3).

Note 4 Peaking suitability of large hydropower plants

Since hydropower plants differ in terms of peaking capabilities, operational strategies must be selected with care. If hydropower plants are operated at a too high C_{OR} , the corresponding peaks in flexible water release may exceed the maximum turbine throughput on a structural basis (cf. equation (S17)), meaning that the plant will be structurally unable to meet peak demand as well as leading to loss of spinning reserves (see Note 8). Typically, this would first occur in the seasons with highest peak demand and/or when water levels are at their lowest. Therefore, each hydropower plant's operation should happen at an optimised value $C_{OR} = C_{OR}^{opt}$ that ensures this is not the case, while maintaining adequate levels of flexibility. In REVUB, the default procedure (which can be changed by the user in the code) is to define the turbine utilisation rate $k_{turb}(t)$ at each time step:

$$k_{turb}(t) = \frac{Q_{stable}(t) + Q_{flexible}(t)}{Q_{turb}^{max}}, \quad (S28)$$

and define hydropower plant operation (at given C_{OR}) as unsuitable for peaking purposes when $p_{99}(k_{turb})$, i.e. the 99th percentile of k_{turb} , is unity (in other words, when turbine capacity is fully exhausted in at least one out of every 100 hours).

As default for each hydropower plant, REVUB uses $C_{OR}^{opt} = C_{OR}^{max}$, the maximum allowed operational flexibility. If operation is found to be unsuitable for peaking purposes at C_{OR}^{max} , the REVUB code automatically resimulates with incrementally reduced C_{OR} , until a value C_{OR}^{opt} is identified for which $p_{99}(k_{turb}) < 1$.

Note 5 Reservoir simulation for small hydropower plants

To assess the balancing potential of small hydropower plants (those with less than a year of storage), which are in reality much more ubiquitous than large hydropower plants, we define the alternative filling time $\tau_{fill,frac}$ corresponding to the amount of years it would take for a fixed fraction of the incoming flow, denoted $Q_{in,frac}(t)$, to fill the reservoir:

$$\tau_{fill,frac} = \frac{V_{max}/T_{year}}{Q_{in,frac}(t)}, \quad (S29)$$

where $Q_{in,frac}(t)$ is defined as

$$Q_{in,frac}(t) = f_{reg} Q_{in,nat}(t), \quad (S30)$$

with f_{reg} a suitable fraction, representing the fraction of incoming water available for storage and, therefore, flexible use. This fraction can be input in REVUB by the user. In case the user is unsure of a pertinent value, it can be left empty and REVUB defaults to a standard value determined by solving $\tau_{fill,frac} = 1$; that is, f_{reg} then represents the fraction of the incoming water that would take one year to fill the reservoir on average. Note that this default is very realistic for hydropower plants on rivers with extremely seasonal, unimodal discharge, but not necessarily as useful in bimodal-rainfall climates or in situations with relatively flat inflow profiles (e.g. in cases where a regulating dam is already present upstream).

Note that f_{reg} would normally be unity for the large (more-than-a-year storage) plants described in the previous sections; nevertheless, the user is free to specify a value smaller than unity for such plants when running REVUB. This could, for instance, reflect a requirement for a seasonal environmental flow, or a safeguard against extremely dry years to prevent regular operating rules from overdrawing the reservoir. In such a case, the operation of those plants, too, will follow what is described below, rather than what is described in the previous sections.

Once f_{reg} chosen, the fraction $Q_{in,frac}(t)$ is then assumed to be “storable” and, after accounting for the imposed environmental and irrigation outflow $Q_{stable}^{env/irr}(t)$, usable for balancing. REVUB assumes that the remainder of the flow, given by $Q_{in,nat}(t) - Q_{in,frac}(t) = (1 - f_{reg})Q_{in,nat}(t)$, passes through the reservoir without being stored, thus representing the seasonal “run-of-river” component $Q_{RoR}(t)$. Since f_{reg} can be calculated based on long-term average flow, such operation can be readily implemented as long as accurate measurements of inflow are available.

For $f_{reg} < 1$, d_{min} no longer represents the fraction of yearly average inflow required as minimum stable outflow, but the fraction of yearly average *storable* flow required as minimum stable outflow. The user must take this into account when specifying d_{min} . REVUB does this automatically when calculating a default d_{min} in case the user does not wish to prescribe a value^{xi}. The term Q_{RoR} is appended in equation (S3) (and would be an additional entry under the “water releases” arrows in Fig. S2):

$$\begin{aligned} V(t + \Delta t) &= V(t) + \left[Q_{in,nat}(t) - Q_{RoR}(t) - Q_{stable}(t) - Q_{flexible}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)] \frac{A(t)}{\rho} \right] \Delta t \\ &= V(t) + \left[Q_{in,frac}(t) - Q_{stable}(t) - Q_{flexible}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)] \frac{A(t)}{\rho} \right] \Delta t, \end{aligned} \quad (S31)$$

and the corresponding seasonal power generation is then equal to

$$P_{RoR}^{hydro}(t) = \min \left[Q_{RoR}(t); \max \left[0; Q_{turb}^{max} - Q_{stable}(t) - Q_{flexible}(t) \right] \right] \eta_{turb} \rho g h(t). \quad (S32)$$

Electricity generation from the hydropower component, denoted E_{RoR}^{hydro} , is then equal to

$$E_{RoR}^{hydro} = \int_{year} P_{RoR}^{hydro}(t) dt. \quad (S33)$$

Dispatching such as described for large hydropower plants in the previous sections is done similarly for small hydropower plants, with the following adaptations: (i) the component $Q_{in,frac}(t)$ is used as reference flow instead of $Q_{in,nat}(t)$ in equation (S4); (ii) the rule is added that, as soon as reservoir levels drop precariously low (in this case, below $f_{restart} V_{max}$), the seasonal outflow (if nonzero) is temporarily stopped and the operation reverts to that of large hydropower plants, using the full inflow to stabilise water levels; and (iii) the seasonal power generation $P_{RoR}^{hydro}(t)$ is not included in equation (S9) as it is not expected to take part in the hour-to-hour load-following. Note, however, that if river discharge would be regulated by other reservoirs upstream in a cascade-like configuration, even $P_{RoR}^{hydro}(t)$ could become relatively stable throughout the year. Note that, for extremely seasonal rivers, it is possible that the run-of-river component of discharge and power generation, after accounting for the stable and flexible components, fully “maxes out” the turbine capacity. This would normally only happen during the periods in which the river discharge reaches its natural peak. In those periods, the plant automatically acts as baseload provider (using 100% of its discharge capacity and/or rated power capacity) and only reverts to the flexible operation once the inflow has somewhat reduced, freeing up operational reserves.

Note 6 Optional pumped-storage assessment

The REVUB code can also be used to assess the potential for increasing load-following potential by refurbishing large hydropower plants to pumped-storage schemes¹⁷, such that excess production (typically of solar PV during mid-day) can be used to pump water back up into the reservoir, enabling the energy to be stored for use at a later stage. This would involve (i) building a downstream reservoir with volume $V_{lower,max} \ll V_{max}$ to store part of the released water, from which it can be pumped back up into the reservoir, and (ii) ensuring the turbines can be operated in reverse mode and/or pumps are installed.

The hydrological balance components then change as follows. First, since environmental and irrigation flow constraints still have to be met, the component $Q_{stable}^{env/irr}(t)$ has to be directly discharged downstream and cannot be used for filling the lower reservoir. Only the component $Q_{flexible}(t)$ and

^{xi}The calculation of this default, in a more generalised form for any f_{reg} , is as follows: $d_{min}^{default} = (f_{min} Q_{turb}^{max} / N_{turbines} - (1 - f_{reg}) \min [Q_{in,nat}(t)]) / (f_{reg} \overline{Q_{in,nat}(t)})$. The term $(1 - f_{reg}) \min [Q_{in,nat}(t)]$ represents the fact that the run-of-river component can contribute to the minimum load of the first turbine provided that it has a nonzero minimum throughout the year. For $f_{reg} = 1$, which is the case for “large” hydropower plants, this equation reduces to $d_{min}^{default} = (f_{min} Q_{turb}^{max} / N_{turbines}) / \overline{Q_{in,nat}(t)}$ as given earlier.

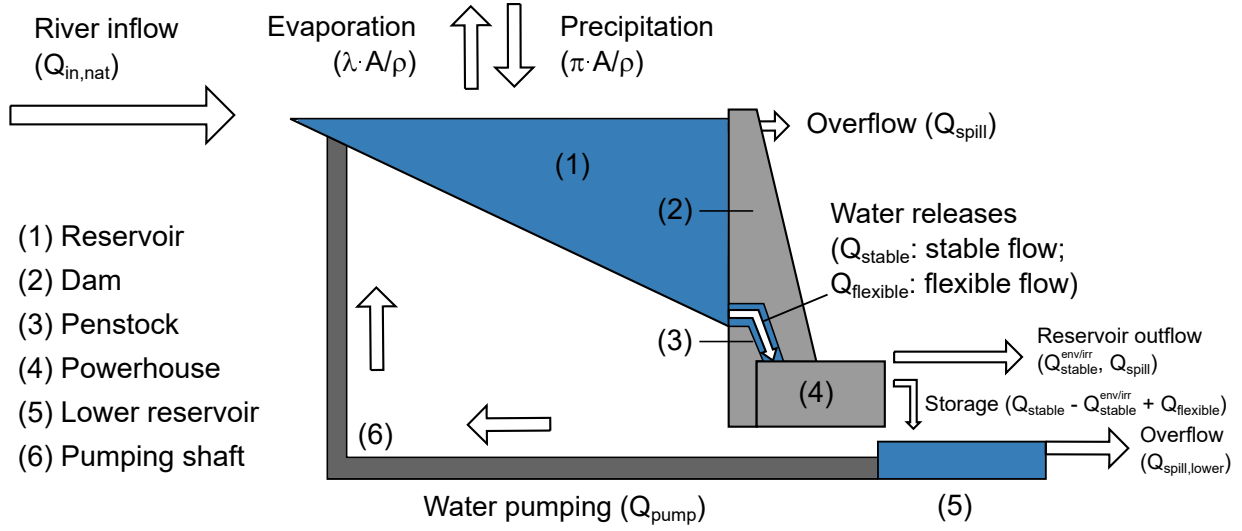


Figure S3: **Flowchart of the water gains and losses of a large hydropower plant, including pumped storage.** Water gains and losses of a hydropower plant modelled as pumped-storage plant, with an additional lower reservoir and a flow of water in the upstream direction for temporary energy storage.

what remains of $Q_{stable}^{tech}(t)$, i.e. the part unaccounted for by $Q_{stable}^{env/irr}(t)$, can be used for filling the lower reservoir. Thus, the water balance of the lower reservoir is then given by

$$\begin{aligned} V_{lower}(t + \Delta t) &= V_{lower}(t) + [Q_{stable}(t) - Q_{stable}^{env/irr}(t) + Q_{flexible}(t) - Q_{pump}(t) - Q_{spill,lower}(t)] \Delta t \\ &= V_{lower}(t) + [\max[Q_{stable}^{tech}(t), Q_{stable}^{env/irr}(t)] - Q_{stable}^{env/irr}(t) + Q_{flexible}(t) - Q_{pump}(t) - Q_{spill,lower}(t)] \Delta t \\ &\equiv V_{lower}(t) + [Q_{in,lower}(t) - Q_{pump}(t) - Q_{spill,lower}(t)] \Delta t, \end{aligned} \quad (S34)$$

where $Q_{pump}(t)$ is the water pumped back up to the large reservoir for storage, $Q_{spill,lower}(t)$ is the overflow component of the lower reservoir, and $Q_{in,lower}(t)$ has been defined as the sum of the inflow into the lower reservoir from the upper reservoir for ease of notation. The precipitation and evaporation terms are neglected here as the lower reservoir is assumed to have a much smaller surface area than the upper reservoir. The water balance of the large reservoir is then given (cf. equation (S3)) by

$$V(t + \Delta t) = V(t) + \left[Q_{in,nat}(t) - Q_{stable}(t) - Q_{flexible}(t) + Q_{pump}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)] \frac{A(t)}{\rho} \right] \Delta t, \quad (S35)$$

and the net outflow of the pumped-storage plant is given (cf. equation (S2)) by

$$Q_{out}(t) = Q_{stable}^{env/irr}(t) + Q_{spill}(t) + Q_{spill,lower}(t). \quad (S36)$$

A schematic of this operation is shown in Fig. S3 (cf. Fig. S2). Note that this kind of pumped-storage operation is mostly applicable for large hydropower plants, as small plants have to continuously discharge during large parts of the year to prevent spillage, and pumping water back up would thus not serve a purpose.

The component $Q_{pump}(t)$ is calculated as follows. The power stored by pumping is given by

$$P_{pump}(t) = \begin{cases} 0, & \text{for } P_d(t) < 0 \\ \min \left[Q_{pump}^{pot}(t) \eta_{pump}^{-1} \rho g h(t); \min[P_d(t); P_{pump}(t-1) + \Delta P_{pump}^{ramp}] \right], & \text{for } P_d(t) \geq 0 \text{ \& } \Delta P_d(t) \geq 0 \\ \min \left[Q_{pump}^{pot}(t) \eta_{pump}^{-1} \rho g h(t); \max[P_d(t); P_{pump}(t-1) - \Delta P_{pump}^{ramp}] \right], & \text{for } P_d(t) \geq 0 \text{ \& } \Delta P_d(t) < 0 \end{cases}, \quad (S37)$$

where ΔP_{pump}^{ramp} is the maximum ramp rate for pumping, η_{pump} is the pumping efficiency, and

$$Q_{pump}^{pot}(t) = \min \left[\frac{V_{lower}(t)}{\Delta t}; Q_{pump}^{max} \right] \quad (S38)$$

with Q_{pump}^{max} the maximum pump throughput. The flow pumped back into the reservoir then equals

$$Q_{pump}(t) = \frac{P_{pump}(t)}{\eta_{pump}^{-1} \rho g h(t)}. \quad (S39)$$

Finally, the overflow component of the lower reservoir is calculated as

$$Q_{spill,lower}(t) = \begin{cases} 0, & \text{for } (V_{lower,max} - V_{lower}(t)) / \Delta t \geq Q_{in,lower}(t) \\ Q_{in,lower}(t) - (V_{lower,max} - V_{lower}(t)) / \Delta t, & \text{for } (V_{lower,max} - V_{lower}(t)) / \Delta t < Q_{in,lower}(t) \end{cases} \quad (S40)$$

Since the pumping allows a certain fraction of the water to be recycled for power generation, the values c_{solar}^{opt} and c_{wind}^{opt} resulting from the optimisation described in Note 3.2 will end up higher as compared to a situation without pumping component. In this way, pumped storage can help increase the load that can be carried by the hydro-solar-wind combination.

The REVUB code includes an optional section to perform the above operations and the required optimisation (this scenario is named “STOR” in the code, as opposed to “CONV” and “BAL”).

Note 7 Modelling cascade plants

The REVUB model includes provisions to allow modelling of cascaded plants, where two or more plants directly downstream of each other interact. In particular, REVUB allows for the following situations to be modelled:

- **A run-of-river-plant is located directly downstream of a reservoir plant.** In this case, the run-of-river plant needs only a very reduced amount of data input: installed capacity, hydraulic head, design discharge, number of turbines, and turbine efficiency. The upstream reservoir plant must be modelled with all the required parameters mentioned in the previous sections, and additionally it must be indicated that the upstream reservoir plant feeds the downstream one. In this way, the downstream plant will take the modelled outflow of the upstream one as inflow data, plus any unregulated lateral inflow. The CONV outflow from the upstream plant is used for this end^{xii}, multiplied with a boolean indicating whether the upstream reservoir plant was active in each hour (i.e. the assumption is that shutdowns upstream lead to shutdowns of the downstream plant too).
- **A reservoir plant is located directly downstream of another reservoir plant; the upstream one is the main flexibility provider.** In this case, it is assumed that the operation of the upstream one is optimised based on its own reservoir capacity, and that the downstream one simply receives the resulting outflow of the upstream one and uses that for its own storage operation. The calculation is done as in the previous point, except that the downstream reservoir plant evidently needs the full set of parameters normally required for a REVUB simulation.
- **A reservoir plant is located directly downstream of another reservoir plant; the downstream one is the main flexibility provider.** In this case, it is assumed that the downstream plant uses the storage capacity of *both* reservoirs, denoted V_{down} (for the downstream plant) and V_{up} (for the upstream plant), respectively, to optimise its operation. Full sets of data for both plants must be entered, and the calculation is done as follows:
 1. The simplified assumption is made that any change ΔV in cumulatively stored volume $V_{cumul}(t) = V_{down}(t) + V_{up}(t)$ across the reservoirs is proportionally divided over both, according to shares f_{down} and f_{up} that the user can determine, with $f_{down} + f_{up} = 1$. In case of no specification by the user, this share automatically defaults to each reservoir’s share in maximum total volume, i.e. $f_{down/up} = V_{down/up}^{max} / V_{cumul}^{max}$. Thus, if a total amount ΔV of previously stored water is turbinised by the downstream plant, it is assumed that this results in a change in storage of $f_{down}\Delta V$ in the downstream reservoir, and $f_{up}\Delta V$ in the upstream reservoir.
 2. First, the simulation of the downstream reservoir plant is run. Any calculations involving overall storage volume, such as eq. (S3), are done using $V_{cumul}(t)$, and not $V_{down}(t)$, as basis.

^{xii}The logic for using CONV being that BAL outflow typically fluctuates strongly from hour to hour, but most run-of-river plants have some amount of “pondage” allowing to smoothen out such fluctuations.

Based on the above logic, the amount of water stored in only the downstream reservoir is calculated as $V_{down}(t) = f_{down} V_{cumul}(t)$, and used to calculate the hydraulic head $h_{down}(t)$ at any point in time, needed to calculate power output of the downstream reservoir. For calculating evaporation and precipitation losses/gains, the surface area of both reservoirs is added to form $A_{cumul}(t) = A_{down}(t) + A_{up}(t)$, and any calculations involving overall area, such as again eq. (S3), are done with $A_{cumul}(t)$, and not $A_{down}(t)$, as basis. The regulation fraction $f_{reg,cumul}$, applied to operation of cumulative storage in the cascade, is interpreted to represent the fraction of total flow into the downstream reservoir ($Q_{in,down}(t)$) allocated for storage across the total available volume (V_{cumul}^{max})^{xiii}. Lastly, the critical level of the “cumulative” storage is taken to be the maximum of $f_{stop,down}$ and $f_{stop,up}$ (and idem for $f_{restart}$).

3. Second, the simulation of the upstream reservoir plant is run. Since its operation is dictated by the needs of the downstream plant (cf. the previous point), the calibration of its volume levels is slightly changed as compared to equation (S20): instead of using $V_{CONV}(t)$ as calibration series, we use the volume curve for the upper reservoir implied by the previous point, i.e. $V_{BAL,up}(t) = f_{up} V_{BAL,cumul}(t)$. The f_{reg} parameter is imposed on the upstream reservoir, based on the value of the same parameter for the cumulative storage (see above) and the ratio of respective average inflows: $f_{reg,up} = f_{reg,cumul} \cdot f_{up} \cdot \overline{Q_{in,down}} / \overline{Q_{in,up}}$.

Since the REVUB model does not currently have detailed routing procedures for cascades with reservoirs that lie relatively far apart, the application of the cascade module is currently limited to situations where downstream flow peak attenuation and flow delay between upstream and downstream reservoirs are small to negligible, i.e. where the reservoirs lie (more or less) directly downstream of one another.

Note 8 Meeting spinning reserve requirements

In electricity networks, a certain reserve capacity, to be utilised in case of disruptions of supply, must always be available. The reserve capacity that is already online is denoted the spinning reserve, of which hydropower plants with reservoirs are typical providers. However, the flexible hydropower operation modeled by REVUB requires hydropower output to be increased during certain intervals to compensate for reductions in solar and/or wind power; thus, the remaining spinning reserves available to manage contingencies will consistently reach minimum values during those times.

The question is thus whether the hydropower plants could still respond to contingency events (e.g. failure of a transmission line leading to a solar power park temporarily being disconnected), despite being operated by flexible rule curves that lead to regular minima in the available spinning reserves. The spinning reserves $P_{spin}(t)$ from all hydropower plants within a certain geography can be calculated as

$$P_{spin}(t) = \sum_{\substack{\text{all hydro} \\ \text{plants}}} [1 - CF_{hydro}(t)] P_{turb}^r \Gamma(t), \quad (S41)$$

where $\Gamma(t)$ equals zero if the hydropower plant has temporarily undergone a drought-related shutdown, and unity otherwise; and $CF_{hydro}(t)$ is the capacity factor of each hydropower plant, calculated as

$$CF_{hydro}(t) = \frac{P_{stable}^{hydro}(t) + P_{flexible}^{hydro}(t) + P_{RoR}^{hydro}(t)}{P_{turb}^r}. \quad (S42)$$

To test the ability of hydropower to respond to supply disruptions of the added solar and wind power, the user should (ex-post) calculate the ratio (denoted ν) of $P_{spin}(t)$ to the total solar and wind power generation assumed to feed into the same grid. Typical benchmarks of spinning reserve needs for high-RE systems are in the order of $\nu \sim 10 - 20\%$ ²⁰.

^{xiii}This assumes, implicitly, that the order of magnitude of any lateral flows feeding the downstream reservoir (i.e. flows which do not pass the upper reservoir) does not exceed the order of magnitude of the storage capacity of the downstream reservoir, which generally seems a reasonable assumption.

440 The table below provides an overview of all described parameters used in the REVUB model and
 441 its inputs and outputs. The symbol of each parameter is provided alongside its unit, a description
 442 of what it represents in the model, its data type (constant, variable, time series), its use in the
 443 model (as input, output, intermediate result, or used for preparing input), and its applicability across
 444 various components of the model. Here, “scenario” denotes a certain simulation setting: different
 445 scenarios may have different solar/wind mixes, different constraints on overproduction, different pools
 446 of hydropower plants contributing to flexibility, etc.

Symbol	Unit	Description	Type	Used as/for	Applicable to
$A(t)$	m^2	Reservoir lake surface area	Time series	Output	each hydropower plant & each scenario
A_{max}	m^2	Maximum reservoir lake surface area (reached at maximum filling level V_{max})	Constant	Input	each hydropower plant
$A_{down}(t)$	m^2	Estimated surface area of water of <i>downstream</i> reservoir in a cascade	Time series	Output	each cascade & each scenario
$A_{up}(t)$	m^2	Estimated surface area of water of <i>upstream</i> reservoir in a cascade	Time series	Output	each cascade & each scenario
$V_{cumul}(t)$	m^3	Cumulative surface area of water in reservoirs in a cascade. Equal to $A_{down}(t) + A_{up}(t)$	Time series	Output	each cascade & each scenario
C_{OR}	-	Operating reserve coefficient: Fraction of yearly reservoir inflow available for flexible use	Variable	-	each hydropower plant
C_{OR}^{max}	-	Maximum allowed operating reserve coefficient taking into account environmental flow needs	Constant	Input	each hydropower plant
C_{OR}^{opt}	-	Optimal operating reserve coefficient	Constant	Output	each hydropower plant & each scenario
c_{solar}	MW	Installed capacity of solar panels	Constant	Output	each hydropower plant & each scenario
c_{solar}^{opt}	MW	Optimal installed solar power capacity needed to follow optimal load $L_{opt}(t)$	Constant	Output	each hydropower plant & each scenario
c_{wind}	MW	Installed capacity of wind turbines	Constant	Output	each hydropower plant & each scenario
c_{wind}^{opt}	MW	Optimal installed wind power capacity needed to follow optimal load $L_{opt}(t)$	Constant	Output	each hydropower plant & each scenario
$CF_{hydro}(t)$	-	Capacity factor of hydropower plants	Time series	Output	each hydropower plant & each scenario
$CF_{solar}(t)$	-	Weighted average capacity factor of solar panels across locations	Time series	Input	each scenario
$CF_{solar}^n(t)$	-	Capacity factor of solar panels in each location with index n	Time series	Input	each solar power site
$CF_{wind}(t)$	-	Weighted average capacity factor of wind turbines across locations	Time series	Input	each scenario
$CF_{wind}^n(t)$	-	Capacity factor of wind turbines in each location with index n	Time series	Input	each wind power site
d_{min}	-	Fraction of average storable reservoir inflow required as minimum outflow	Constant	Input	each hydropower plant
$d_{min}^{default}$	-	Default value of d_{min} used by REVUB in case of non-specification by user	Constant	Input	each hydropower plant
ΔP_{turb}^{ramp}	MW/min	Maximum ramp rate of hydropower plant	Constant	Input	each hydropower plant
ΔP_{pump}^{ramp}	MW/min	When simulating pumped-storage potential: Maximum ramp rate of pumps	Constant	Input	each large hydropower plant
$E_{reservoir}^{hydro}$	GWh/year	Total yearly hydropower generation	Yearly total	Output	each hydropower plant & each scenario
E_{stable}^{hydro}	GWh/year	Total yearly hydropower generation from stable reservoir outflow component	Yearly total	Output	each hydropower plant & each scenario
$E_{flexible}^{hydro}$	GWh/year	Total yearly hydropower generation from flexible reservoir outflow component	Yearly total	Output	each hydropower plant & each scenario
E_{RoR}^{hydro}	GWh/year	Total yearly hydropower generation from seasonal run-of-river outflow component	Yearly total	Output	each small hydropower plant & each scenario
E_{solar}	GWh/year	Total yearly solar power generation	Yearly total	Output	each hydropower plant & each scenario
E_{wind}	GWh/year	Total yearly wind power generation	Yearly total	Output	each hydropower plant & each scenario
ELCC	GWh/year	Effective Load Carrying Capability. Total yearly load followed by hydro-solar-wind without loss of load, ensuring long-term lake level stability, and meeting environmental flow requirements. Integral of $L_{followed}(t)$ over all time steps in a year.	Yearly total	Output	each hydropower plant & each scenario
ELCC _{tot}	GWh/year	Aggregate of ELCC across all power plants. Also called “total load-following potential” of hydro-solar-wind.	Yearly total	Output	each scenario

Symbol	Unit	Description	Type	Used as/for	Applicable to
η_{turb}	-	Hydroturbine conversion efficiency (note: can be used to include representation of power factor)	Constant	Input	each hydropower plant
η_{pump}	-	When simulating pumped-storage potential: Pumping efficiency	Constant	Input	each hydropower plant
f_{down}	-	Fraction of downstream reservoir volume in cumulative storage volume in a cascade	Constant	Input	Each cascade
f_{up}	-	Fraction of upstream reservoir volume in cumulative storage volume in a cascade	Constant	Input	Each cascade
f_{opt}	-	Optimal filling fraction of hydropower reservoir	Constant	Input	each hydropower plant
f_{spill}	-	Reservoir filling fraction at which overflow prevention via spillways is initiated	Constant	Input	each hydropower plant
f_{stop}	-	Low reservoir filling fraction at which hydropower generation is curtailed	Constant	Input	each hydropower plant
$f_{restart}$	-	Reservoir filling fraction at which hydropower generation is restarted after curtailment	Constant	Input	each hydropower plant
f_{reg}	-	Fraction of yearly average natural inflow allocated to flexible use	Constant	Input	each hydropower plant
g	m/s ²	Gravitational acceleration	Constant	Input	general
γ	-	Conventional reservoir rule parameter	Constant	Input	each hydropower plant
$\Gamma(t)$	-	Binary function indicating whether hydropower plant is operating (1) or not (0)	Time series	Output	each hydropower plant & each scenario
$h(t)$	m	Hydraulic head: Difference in water level between headwater (behind the dam) and tailwater (at the turbines)	Time series	Output	each hydropower plant & each scenario
h_{max}	m	Maximum hydraulic head of hydropower scheme (reached at maximum filling level V_{max})	Constant	Input	each hydropower plant
$k_{turb}(t)$	-	Hydroturbine utilisation rate	Time series	Output	each hydropower plant & each scenario
κ	-	Conventional reservoir rule parameter	Constant	Input	each hydropower plant
$L(t)$	MW	Load to be followed	Time series	Intermediate	each scenario
$L_{opt}(t)$	MW	Optimal load to be followed by hydro-solar-wind mix, ensuring long-term reservoir lake level stability	Time series	Output	each hydropower plant & each scenario
$L_{followed}(t)$	MW	Actual followed load under optimal solution at zero loss of load (ideally equal to $L_{opt}(t)$)	Time series	Output	each hydropower plant & each scenario
$L_{res}(t)$	MW	Residual load: Difference between actual followed load and total hydro-solar-wind power generation	Time series	Intermediate	each hydropower plant & each scenario
$L_{norm}(t)$	-	Normalised load curve shape	Time series	Input	each scenario
$\lambda(t)$	kg/m ² /s	Evaporation flux on reservoir lake surface	Time series	Input	each hydropower plant
μ	-	Constant used in modelling overflow prevention via spillways	Constant	Input	each hydropower plant
n	-	Index for different locations of solar and wind power generation	Index	-	solar power and wind power sites
N_{solar}	-	Total number of solar power generation sites connected to the same power grid	Constant	Pre-input	each scenario
N_{wind}	-	Total number of wind power generation sites connected to the same power grid	Constant	Pre-input	each scenario
$N_{turbines}$	-	Total number of turbines (units) of a single hydropower plant	Constant	Pre-input	each hydropower plant
$\nu(t)$	-	Ratio of spinning reserves to total solar and wind power generation	Time series	Output	each hydropower plant & each scenario
P_{turb}^r	MW	Rated hydroturbine capacity	Constant	Input	each hydropower plant
$P_{stable}^{hydro}(t)$	MW	Power generated by stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$P_{flexible}^{hydro}(t)$	MW	Power generated by flexible component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$P_{RoR}^{hydro}(t)$	MW	Power generated by seasonal run-of-river component of reservoir outflow	Time series	Output	each small hydropower plant & each scenario
$P_{inflexible}(t)$	MW	Inflexible power generation (stable hydro + solar + wind)	Time series	Intermediate	each hydropower plant & each scenario
$P_{solar}(t)$	MW	Solar power generation	Time series	Output	each hydropower plant & each scenario
$P_{wind}(t)$	MW	Wind power generation	Time series	Output	each hydropower plant & each scenario
$P_{followed}$	MW	Average actual followed load under optimal solution	Constant	Output	each hydropower plant & each scenario
$P_{guaranteed}^{xx}$	MW	Guaranteed power (based on user-defined exceedance percentile xx) across simulation horizon	Constant	Output	each hydropower plant & each scenario
P_{load}	MW	Average load to be followed	Constant	Intermediate	each hydropower plant & each scenario

Symbol	Unit	Description	Type	Used as/for	Applicable to
$P_{spin}(t)$	MW	Spinning reserves available from all hydropower plants	Time series	Output	each hydropower plant & each scenario
$P_{pump}(t)$	MW	When simulating pumped-storage potential: Power used for pumping	Time series	Output	each large hydropower plant & each scenario
$P_d(t)$	MW	Difference between total inflexible power generation (stable hydro + solar + wind) and to-be-followed load	Time series	Output	each hydropower plant & each scenario
ϕ	-	Conventional reservoir rule parameter	Constant	Input	each hydropower plant
$\pi(t)$	kg/m ² /s	Precipitation flux on reservoir lake surface	Time series	Input	each hydropower plant
Ψ	-	Relative deviation between simulated lake levels under balancing-oriented and conventional operation, respectively	Function of c_{solar}, c_{wind}	Intermediate	each hydropower plant & each scenario
$Q_{in,frac}(t)$	m ³ /s	Part of natural inflow assumed to be storable, equal to f_{reg} times the natural inflow	Time series	Intermediate	each small hydropower plant
$Q_{in,nat}(t)$	m ³ /s	Natural river discharge into reservoir	Time series	Input	each hydropower plant
$Q_{in,lower}(t)$	m ³ /s	Upper reservoir discharge into lower reservoir for pumped-storage plants, consisting of the non-environmental/irrigation component of stable outflow as well as flexible outflow	Time series	Input	each hydropower plant & each scenario
$Q_{out}(t)$	m ³ /s	Total reservoir outflow, consisting of stable, flexible, run-of-river and spilling components	Time series	Output	each hydropower plant & each scenario
$Q_{stable}(t)$	m ³ /s	Stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q_{stable}^{tech}(t)$	m ³ /s	Technologically required part (related to minimum stable turbine load and need for active synchronous capacity) of stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q_{stable}^{env/irr}(t)$	m ³ /s	Environmentally/irrigation-wise required part of stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q_{flexible}(t)$	m ³ /s	Flexible component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q_{RoR}(t)$	m ³ /s	Seasonal run-of-river component of reservoir outflow (zero for large hydropower plants)	Time series	Output	each small hydropower plant
$Q_{spill}(t)$	m ³ /s	Spilling component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
Q_{turb}^{max}	m ³ /s	Maximum hydroturbine throughput	Constant	Input	each hydropower plant
$Q_{turb,flexible}^{pot}(t)$	m ³ /s	Maximum potential outflow available for flexible hydropower generation, before accounting for needs and ramping constraints	Time series	Intermediate	each hydropower plant & each scenario
$Q_{pump}(t)$	m ³ /s	When simulating pumped-storage potential: Pumped flow	Time series	Output	each large hydropower plant & each scenario
Q_{pump}^{max}	m ³ /s	When simulating pumped-storage potential: Maximum pumping throughput	Constant	Input	each large hydropower plant
ρ	kg/m ³	Density of water	Constant	Input	general
t	hours	Time	Variable	-	general
T_{year}	seconds	Number of seconds in a year	Constant	Input	general
τ_{fill}	-	Filling time: number of years it takes (on average) to fill reservoir with natural inflow	Constant	Intermediate	each hydropower plant
$\tau_{fill,frac}$	-	Filling time: number of years it takes (on average) to fill reservoir with fraction f_{reg} of natural inflow	Constant	Intermediate	each small hydropower plant
$V(t)$	m ³	Volume of water in reservoir	Time series	Output	each hydropower plant & each scenario
V_{max}	m ³	Maximum reservoir volume	Constant	Input	each hydropower plant
$V_{down}(t)$	m ³	Estimated volume of water in <i>downstream</i> reservoir in a cascade	Time series	Output	each cascade & each scenario
$V_{up}(t)$	m ³	Estimated volume of water in <i>upstream</i> reservoir in a cascade	Time series	Output	each cascade & each scenario
$V_{cumul}(t)$	m ³	Cumulative volume of water in reservoirs in a cascade. Equal to $V_{down}(t) + V_{up}(t)$	Time series	Output	each cascade & each scenario
$V_{lower}(t)$	m ³	When simulating pumped-storage potential: Volume of lower reservoir	Time series	Output	each large hydropower plant & each scenario
$V_{lower,max}$	m ³	When simulating pumped-storage potential: Maximum volume of lower reservoir	Constant	Input	each large hydropower plant
w_{solar}^n	-	Weight factor for capacity in each solar power site	Constant	Pre-input	each solar power site
w_{wind}^n	-	Weight factor for capacity in each wind power site	Constant	Pre-input	each wind power site
z_{max}	m	Maximal reservoir depth	Constant	Pre-input	each hydropower plant

Note 10 Running the model: Input (A)

The initialisation of the REVUB simulation is done through the file `A_REVUB.initialise`. All parameters to be defined by the user can be controlled using the Excel file `parameters.simulation`, hereafter the “master file”. This includes various types of data:

- *Hydropower-plant specific parameters*, which are constants specific to each hydropower plant. These quantities are set in the worksheet “Hydropower plant parameters” of the master file. In this worksheet, it is also possible to turn on/off individual hydropower plants for simulation inclusion (parameter `HPP_active`); thus, the worksheet can serve as overall database to collect hydropower plant data.
- *Bathymetric relationship* for the modelled hydropower plants. These values should be read in by the code as an array with three columns and a user-determined number of rows; the first column should represent reservoir volume, and the second and third column the corresponding area and hydraulic head values. The relationships are set in the Excel file `data.bathymetry` and linked to the worksheet “Hydropower plant parameters” of the master file. This linking is done through the parameter `HPP_name_data_bathymetry` on that worksheet.
- *Time series* of important parameters whose value needs to be known for each time step before a simulation can be performed, namely reservoir inflow, evaporation, precipitation, solar/wind power capacity factor, and the shape of the electricity demand profile. These values should be read in by the code as two-dimensional matrices for each hydropower plant, with the following dimensions^{xiv}: *[number of time slices (default: hours) per year; number of years covered by the simulation]*.^{xv} These quantities are set in the other Excel files `data.xxx.xlsx` and linked to the worksheet “Hydropower plant parameters” of the master file. This linking is done through the parameters `HPP_name_data_xxx` in that worksheet.
 - ◊ Note that hydrological time series (inflow, precipitation, evaporation) may often only be available at monthly timescale, not hourly, as opposed to solar/wind capacity factors and load. The REVUB repository contains a monthly-to-hourly data converter code (`rearrange_data_monthly_to_hourly`, under “*data/auxiliary scripts*”) that parses user-specified monthly time series (to be entered in columns in the data collection file `rearrange_data_template.xlsx`, sheet `monthly_series`) into the matrix format required for REVUB. Running the converter code provides a matrix named `output_hourly_byyear` that can be copy-pasted into the relevant `data.xxx.xlsx` files.
 - ◊ Similarly, time series for downstream irrigation needs may typically only be available at hourly (or different sub-daily) timescale for separate months of the year. The same may be the case for e.g. simplified solar/wind CF input data in which only the basic sub-daily and seasonal dynamics are represented. The REVUB repository also contains a second data converter code (`rearrange_data_daily_bymonth_to_hourly`, under “*data/auxiliary scripts*”) to parse user-specified 24-h series given separately for each month (to be entered in columns in the file `rearrange_data_template.xlsx`, sheet `daily_bymonth_series`) into the matrix format required for REVUB. Here, too, running the converter code provides a matrix named `output_hourly_byyear` that can be copy-pasted into the relevant `data.xxx.xlsx` files.

^{xiv}Note that these technically refer to minimum dimensions. The user can, for instance, enter data for a higher amount of columns (years) than there are years in the simulation as specified by the time-related parameters above; REVUB will ignore those extra columns. This feature may be used to run initial tests using a limited number of years to speed up simulation time, before running with all years once satisfied with simulations settings.

^{xv}Note that the REVUB code is designed to take into account the extra days occurring in leap years; for the purposes of preparing these data sets, if the simulation period contains leap years, the number of elements along the first dimension of these matrices should thus reflect the number of time slices (hours) in a leap year, and the data for non-leap years should be appended with *nan* values to reach the same length.

- *Indications* of which hydropower plant is in a cascade with which other one, either downstream or upstream. This is done through the parameters `HPP_cascade_upstream` (indicating the name of reservoir plants upstream) and `HPP_cascade_downstream` (indicating the name of reservoir or RoR plants downstream) in the worksheet “Hydropower plant parameters” of the master file.
- *Simulation accuracy parameters* specific to each hydropower plant. These quantities are set in the worksheet “Hydropower plant parameters” of the master file. They are the following:
 - ◊ `f_init_BAL_start`, `f_init_BAL_step` and `f_init_BAL_end` determine the range (start, step size, and end) of $E_{solar} + E_{wind}$ (cf. equation (S25)), expressed as a fraction of $E_{reservoir, CONV}^{hydro}$ (cf. equation (S24)), i.e. the solution space in which the REVUB code starts searching for the solution to equation (S21). Note that the code gives a warning message to the user in case this range turns out to be inadequate (i.e. if no global minimum in Ψ is found). The same ranges should also be given for STOR scenarios (`f_init_STOR_start`, `f_init_STOR_step` and `f_init_STOR_end`) in case the user wishes to model these.
 - ◊ `N_refine_BAL` determines the accuracy with which the solution to equation (S21) is determined. After the initial search for this minimum in the range [`f_init_BAL_start` : `f_init_BAL_step` : `f_init_BAL_end`] (see previous point), the REVUB model can zoom in to the range around this minimum to identify its value with increased accuracy, using a step size reduced by a factor of ten. The number of times with which this is done is determined by `N_refine_BAL`. Thus, `N_refine_BAL` = 1 means that the initial search is deemed accurate enough, whereas each +1 in `N_refine_BAL` increases the accuracy of identification of the minimum in Ψ by one digit. Accordingly, each +1 also increases computation time proportionally. For users wishing to model STOR scenarios, the same principle applies to `N_refine_STOR`.
- *General modelling parameters* not specific to individual hydropower plants. These quantities are set in the worksheet “General parameters” of the master file. They refer to the length of the time series to simulate; to physical, invariant quantities used in the hydropower modelling according to the equations introduced in this Manual; and a few overall simulation accuracy parameters:
 - ◊ `N_ELCC` determines the accuracy (the number of discrete steps between zero and maximum power generation) with which the zero-crossing of L_{res} in equation (S22) is determined.
 - ◊ `psi_min_threshold` is zero by default, but can be set higher to speed up computation time in case the search for the minimum in Ψ takes too long. This parameter ensures that the zooming-in to higher accuracies (see previous point) is stopped as soon as $\min(\Psi)$ ends up below `psi_min_threshold`. It can thus be a compromise for users wishing to have high accuracy of determining $\min(\Psi)$ (i.e. setting `N_refine_BAL/STOR` > 1) but not at a high cost of computational speed.
 - ◊ `X_max_BAL` determines the amount of iterative loops employed by the model to get a converged estimate of $P_{stable, BAL}^{hydro}$ (cf. equation (S9) and the explanation below equation (S19)). Each +1 also increases computation time proportionally. For users wishing to model STOR scenarios, the same principle applies to `X_max_STOR`.

Note 11 Running the model: Core code (B)

The actual core code of REVUB is contained in the file `B_REVUB_main_code`. Once the input data has been loaded using the file `A_REVUB_initialise`, the user can principally run this core code directly.

Note 12 Running the model: Output graphics and tables (C)

The results of the REVUB simulation are post-processed with two separate scripts, which generate several standardised figures and data tables based on the input and output of the model. The two scripts serve a somewhat different purpose:

- The script `C_REVUB_plotting_individual` is used to plot simulation results pertaining to a single (user-selected) hydropower plant, and produce data tables with hourly values of power generation, lake levels, and outflow for CONV and BAL runs of individual plants.
- The script `C_REVUB_plotting_multiple` is used to aggregate simulation results pertaining to (a user-selected set of) all hydropower plants. Based on a user-defined total electricity demand, it shows how much of this total demand can be met by hydro-solar-wind power on all involved timescales, and how much would remain to be met by other power sources. It plots the corresponding simulation results, as well as data tables with hourly values of power generation by source.

Note 12.1 Results for individual power plants

The figures generated by the file `C_REVUB_plotting_individual` are described below.

- Figure 1 shows (a) the (h, V) and (b) the (A, V) bathymetric calibration curves used as input to the model.
- Figure 2 shows (a) the full time series of hydraulic head $h(t)$ under CONV, BAL and (if modelled) STOR, (b) the frequency spectrum of these hydraulic head curves such that e.g. specific temporal signatures, such as diurnal dispatch in solar-heavy systems, can be discerned, and (c) the monthly median and interquartile ranges of natural inflow $Q_{in,nat}(t)$ and outflow $Q_{out}(t)$ (equation (S2)) under CONV, BAL and (if modelled) STOR.
- Figure 3 shows (a) the full time series of lake volume $V(t)$ under CONV, BAL and (if modelled) STOR, and (b) the monthly average time series of natural inflow $Q_{in,nat}(t)$ and outflow $Q_{out}(t)$ under CONV, BAL and (if modelled) STOR.
- Figure 4 shows the monthly average composition of the hydro-solar-wind mix supported by the selected hydropower plant, under the BAL scenario and for a user-defined year (see below). The figure indicates hydropower generation (by stable, flexible and RoR components) and the optimal solar and wind power contribution identified by REVUB. The achieved ELCC, whose profile reflects that of $L_{norm}(t)$, is also indicated. In case a STOR scenario was simulated, a corresponding extra figure will be produced for that scenario, additionally indicating the pump-stored part of solar/wind power generation.
- Figure 5 is analogous to Figure 4, but at yearly resolution and reflecting the entire simulated time series.
- Figure 6 is analogous to Figure 4, but at hourly (full) resolution for a user-defined time slice (see below).
- Figure 7 shows approximated reservoir release rules in the BAL scenario for a user-defined hour of the day during a user-defined month. The plot shows the median and interquartile range of needed reservoir release (minus the RoR and spill components, i.e. $Q_{stable} + Q_{flexible}$) versus the median hydraulic head, with each data point denoting results from one simulation year. A linear fit to these data points is also shown.
- Figure 8 shows statistics of turbine use over the simulated period. Based on the number of units installed (part of the input in the worksheet “Hydropower plant parameters” of the Excel file *parameters_simulation.xlsx*; see above), the amount of time in which a specific number of those is active is extracted from the hourly profiles of power generation and plotted in a bar chart.

- Figure 9 shows statistics of the operational regime in the BAL scenario, divided into four options: flexibility (the regular operation modeled with REVUB), baseload (when the turbine capacity is fully maxed out; see Note 5), mixed (the border regime where the plant oscillates between the flexibility regime and baseload regime; this occurs when total outflow oscillates near the maximum discharge capacity of the turbines), and curtailed (when droughts force hydropower curtailment).

To produce the figures, a few parameters need to be user-defined in the worksheets “Plot power output (single HPP)” of the file *parameters_simulation.xlsx*, where they can be changed to adapt the data to be plotted. These are as follows:

- `plot_HPP` is the index of the hydropower plant for which to plot results in all Figures.
- `plot_year` is the index of the simulation year for which to plot results in Figures 4 and 6.
- `plot_month` is the month in which to start the time slice in Figure 6.
- `plot_day_month` is the day of the month defined by `plot_month` on which to start the time slice in Figure 6.
- `plot_num_days` is the number of days for which to plot results in Figure 6.
- `plot_rules_month` is the month(s) of the year for which to plot results in Figure 7. This can be an array, e.g. [1, 4] would denote that the curves in Figure 7 are plotted both for January and for April.
- `plot_rules_hr` is the hour(s) of day during the month defined by `plot_rules_month` for which to plot results in Figure 7. This can be an array, e.g. [8, 20] would denote that the curves in Figure 7 are plotted both for 8 a.m. and 8 p.m.
- The user can further choose to include or exclude the RoR-component of hydropower in the graphs (`plot_RoR_part`), to include or exclude the plotting of the ELCC as a line (`plot_ELCC_line`), to turn the production of each individual figure on and off (`FigureX_on`), and to turn the production of data table output on and off (`tables_on`).

Note 12.2 Results for multiple power plants

The file `C.REVUB_plotting_multiple` serves to show to what extent the investigated hydropower, solar power and wind power plants could contribute to the overall power mix of a given territory. The principle of this file is as follows:

- To run the file, the user must first define several parameters to construct an overall electricity demand curve, denoted `P_total_hourly` (see below).
- The file then plots the aggregate power generation from all investigated hydro, solar and wind plants under BAL and STOR, and compares it to `P_total_hourly`.
- Any shortfalls in renewable power generation are then assumed to be filled up by thermal power plants, insofar as user-defined available thermal capacity permits. This reflects the assumption of a priority of dispatch for renewables. Any shortfalls unable to be met by this thermal capacity are assumed to be unsatisfied.
- When renewable power generation exceeds `P_total_hourly` by a certain amount, this amount is assumed to be curtailed.

In this way, the file allows to calculate the total power mix (hydro/solar/wind/thermal) suggested by the streamlined hydro-solar-wind operation as simulated by REVUB.

The figures produced by the file are described below.

- Figure 1 shows the monthly average composition of the hydro-solar-wind mix under the BAL scenario for a user-defined year (see below). The figure indicates the total power demand `P_total_hourly`, and how this power demand is met by the aggregate hydropower generation (by stable, flexible and RoR components) from all hydropower plants, the aggregate of all solar and wind power, and the necessary other (thermal) power generation. Any curtailed power (going beyond the user-defined overall demand) or unsatisfied power (unable to be met due to insufficient thermal capacity) is shown as such. In case a STOR scenario was simulated, a corresponding extra figure will be produced for that scenario, additionally indicating the pump-stored part of solar/wind power generation. (In this case, small hydropower plants for which no STOR scenario is available (cf. section Note 6) are included according to their BAL results.)
- Figure 2 is analogous to Figure 1, but at yearly resolution and reflecting the entire simulated time series.
- Figure 3 is analogous to Figure 1, but at hourly (full) resolution for a user-defined time slice (see below).
- Figure 4 shows average ramping envelopes of the hydro and thermal capacity across the same user-defined year as in Figure 1, for a user-defined range of hours. It contains two sub-figures, one of the ramping needs in absolute terms (left) and one of the ramping needs expressed as fraction of total capacity per hour (right).

To produce the figures, a few parameters need to be user-defined in the worksheet “Plot power output (multi HPP)” of the file *parameters_simulation.xlsx*, where they can be changed to adapt the data to be plotted. These are as follows:

- `plot_HPP_multiple` is an array containing the indices of the hydropower plants whose results are to be aggregated in the Figures. The user can thus select all, or a selection of, the simulated hydropower plants.
- `choose_demand_type` is a boolean operator, where 0 should be chosen to use the same (static) demand time series for every year, and 1 signifies the user wishes to show changes (dynamic) in annual demand levels.
- `P_total_av` and `P_r_total_thermal` represent the static values for average power demand and available thermal capacity, respectively, for the case `choose_demand_type = 0`.
- `P_total_av_series` and `P_r_total_thermal_series` represent the dynamic values (one for each year) for average power demand and available thermal capacity, respectively, for the case `choose_demand_type = 1`.
- `plot_year_multiple` is the index of the simulation year for which to plot results in Figures 1 and 3.
- `plot_month_multiple` is the month in which to start the time slice in Figure 3.
- `plot_day_month_multiple` is the day of the month defined by `plot_month` on which to start the time slice in Figure 3.
- `plot_num_days_multiple` is the number of days for which to plot results in Figure 3.
- `plot_ramping_range` is the number of hours across which ramping envelopes should be calculated.
- `chosen_load` is the name of the target load curve to be used in the plotting (found in the Excel file *data_load.xlsx*).
- `plot_ELCC_line_multiple` allows the user to choose to include or exclude the plotting of the ELCC as a line.

- `tables_on_multiple` allows the user to choose to turn the production of output tables with hourly results on and off.

Note 13 Supplementary tips for model use

This section provides a few additional hints that may help to ease the use of the REVUB model in practice.

- As mentioned in sections Note 3.1 and Note 3.2, the default is to use the entire simulation period as reference for several processes in REVUB, such as the calculation and use of long-term average flow statistics (cf. eq. (S4), eq. (S14)) and the optimisation of lake level dynamics of BAL/STOR versus CONV (eq. (S20)). The user can change this by manually setting the parameters `year_calibration_start` and `year_calibration_end` in the sheet “Hydropower plant parameters” of the master file *parameters_simulation.xlsx*. This can be useful, for instance, when simulating extreme events such as unforeseen droughts hitting a hydropower plant: in that case, one would select the calibration period to not include that extreme year, forcing the model to be “myopic” in relation to the extreme dry period. Leaving the parameter fields empty will revert the model to the default of using the entire simulation period.
- In order to check several basic outcomes of the model (such as lake levels, outflow statistics, etc.), it is instructive to first run a CONV scenario without bothering with the BAL/STOR optimisations, allowing to easily identify obvious errors in the input data. For this reason, it is recommended to first run a “calibration” scenario with only CONV by setting `calibration_only` to zero in the worksheet “General parameters” in the master file. This will skip the BAL/STOR optimisation entirely.
- One could conceivably wish to investigate how a hydropower plant performs flexibly (BAL) compared to baseload-like (CONV), but *without* any contribution from solar and/or wind power, i.e. solely to meet a certain target load without hybridisation with other power sources. This can be simulated by simply not linking any solar and wind timeseries to the simulation (leaving `HPP_name_data_CF_solar` and `HPP_name_data_CF_wind` empty in the sheet “Hydropower plant parameters” of the master file), in which case REVUB will run the simulation without solar and wind. In that case, the code uses an alternative for eq. (S11), namely

$$P_{load} = P_{turb}^r \cdot f_{reg} \cdot c_{dummy}, \quad (S43)$$

with c_{dummy} a dummy variable analogous to c_{solar} and c_{wind} from eq. (S9), so that eq. (S21) becomes

$$\min(\Psi) = \Psi(c_{dummy}^{opt}). \quad (S44)$$

- For simplicity, if the user wishes to ignore precipitation and evaporation effects in the simulation (usually second-order effects), they can simply choose to not link any precipitation or evaporation time series to the hydropower plants in question. This is done in the sheet “Hydropower plant parameters” of the master file by leaving the parameters `HPP_name_data_precipitation` and `HPP_name_data_evaporation` empty. The same goes for the prescribed outflow, through the parameter `HPP_name_data_outflow_prescribed`.
- Although REVUB is primarily designed to simulate reservoir hydropower plants, it can be used in “simplified” mode to simulate run-of-river plants. In this case, most of the input parameters can be ignored as they are not relevant to plants without reservoirs. The only remaining parameters of importance are the inflow (`HPP_name_data_inflow` in the sheet “Hydropower plant parameters” of the master file, linking to the correct inflow time series), rated capacity (`P_r_turb`), hydraulic head (`h_max`), design discharge (`Q_max_turb`), number of turbines (`no_turbines`), and turbine efficiency (`eta_turb`). All other fields in the sheet “Hydropower

plant parameters” can be left empty. In case the run-of-river plant is part of a cascade with reservoir plants upstream, even the inflow parameter can be left empty as the plant will receive the outflow of the upstream plant as inflow (cf. Note 7).

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